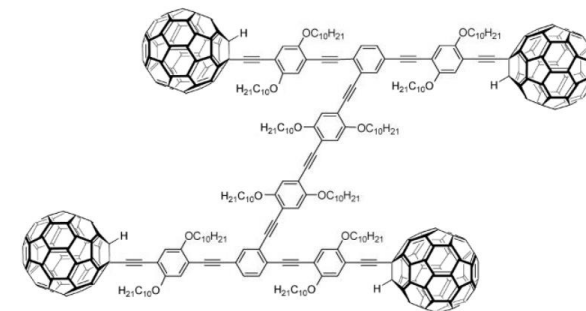


CH223 Organic chemistry II

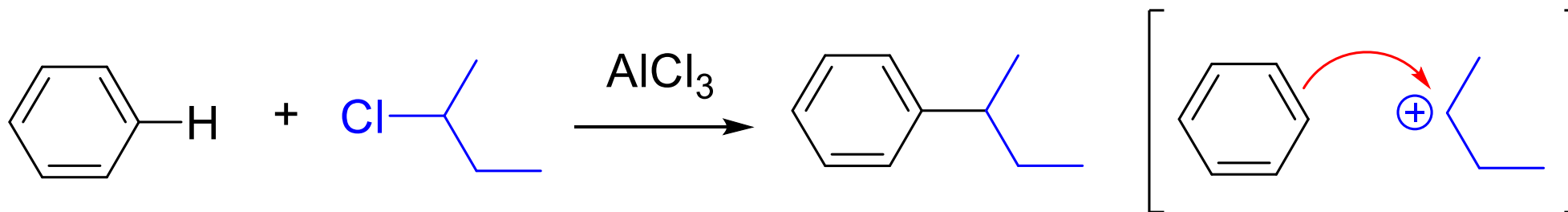
Last class: brief summary of the course

Jun Ohata, Ph.D.

Assistant Professor of Chemistry



Friedel–Crafts Alkylation

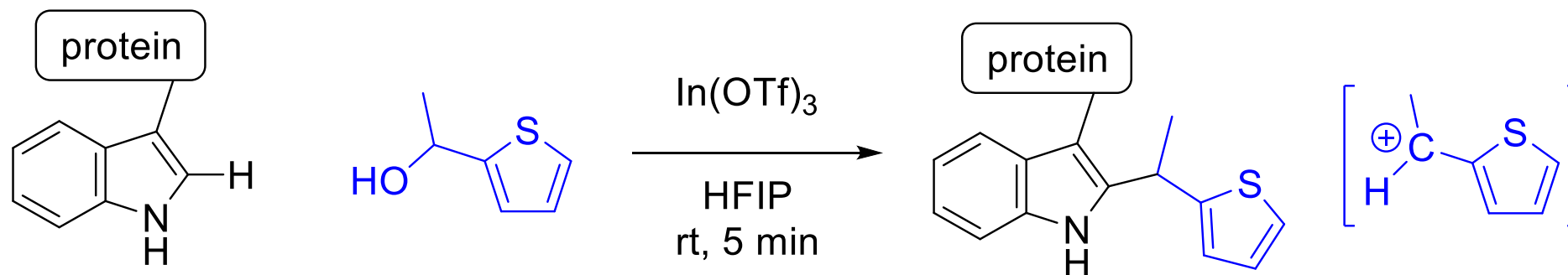


- Alkylation of benzene by electrophilic aromatic substitution.
- Aluminum chloride (AlCl_3) is actually sensitive water (inconvenient to use!)

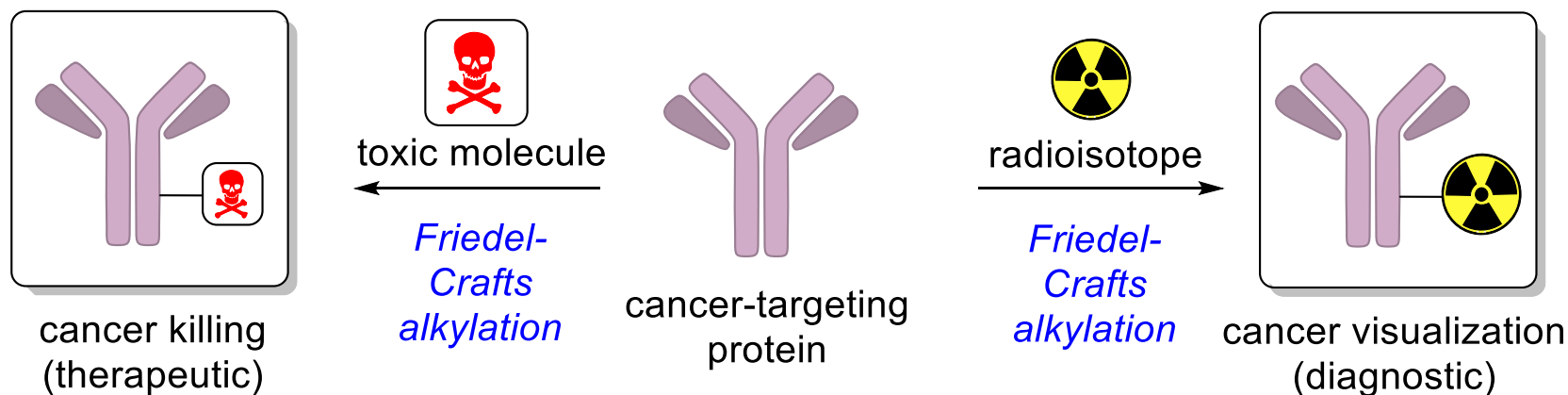


<https://www.youtube.com/watch?v=GIDUYWpJj8g>

Ohata lab: Friedel–Crafts Alkylation for therapeutic and diagnostic applications



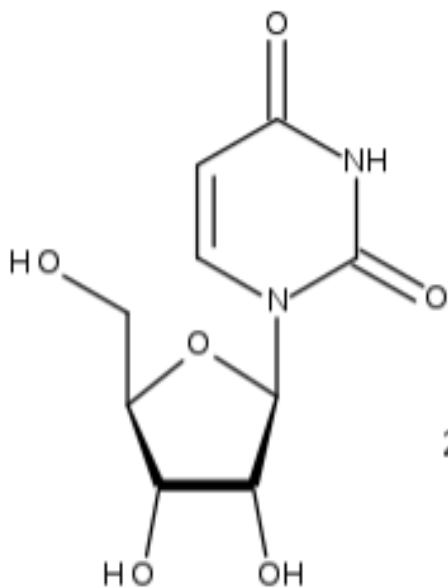
Chiamaka Uzoewulu



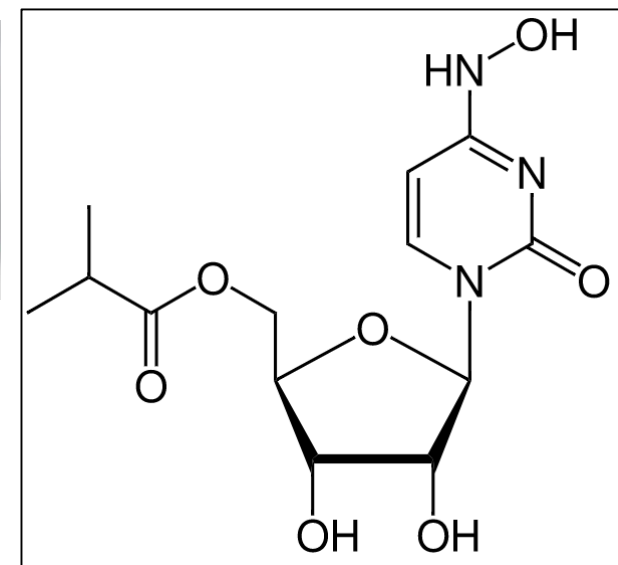
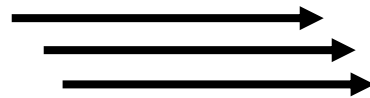
Unpublished results from the Ohata lab

A slide from the first class

How to do this multi-step synthesis?

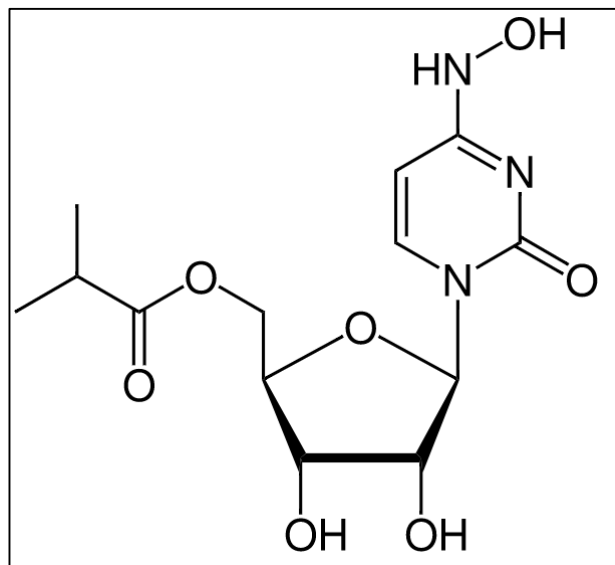


Uridine
(one of the
typical RNA
components)



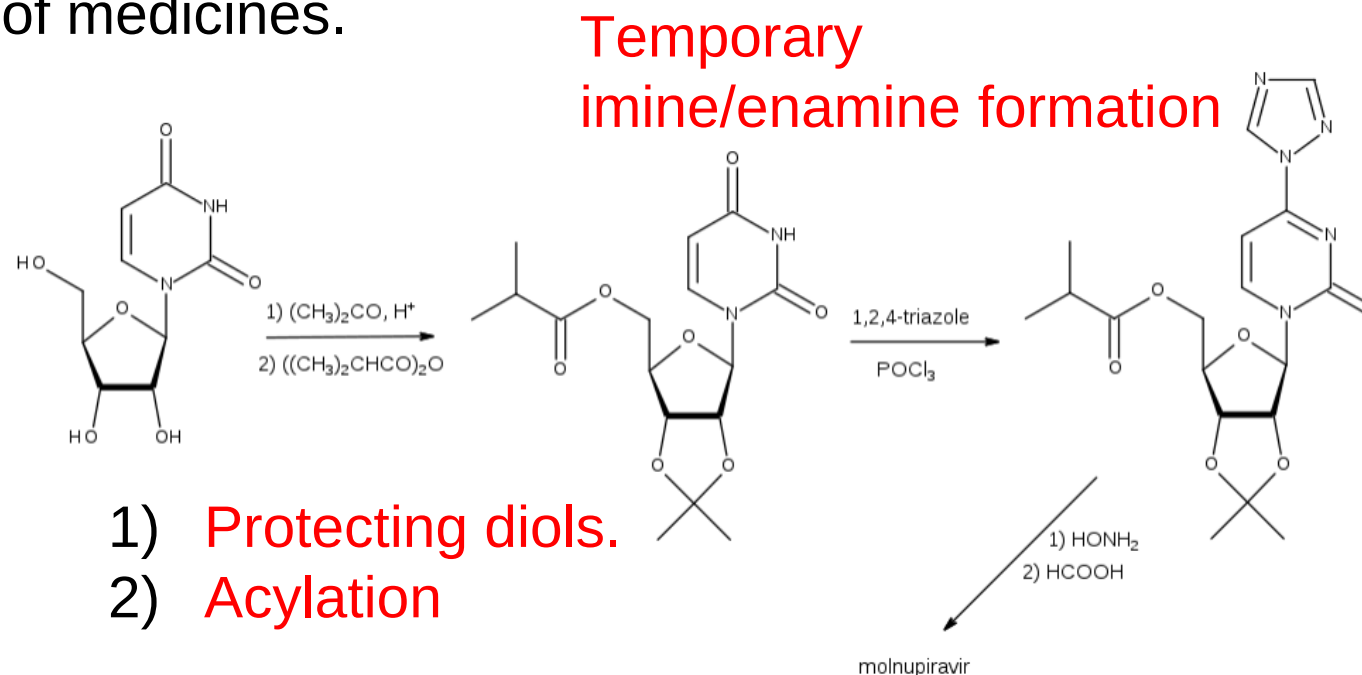
Molnupiravir, sold under the brand name Lagevrio among others, is an antiviral medication that inhibits the replication of certain RNA viruses, and **is used to treat COVID-19 in those infected by SARS-CoV-2.**

Organic compounds and medicines



Molnupiravir, sold under the brand name Lagevrio among others, is an antiviral medication that inhibits the replication of certain RNA viruses, and **is used to treat COVID-19 in those infected by SARS-CoV-2.**

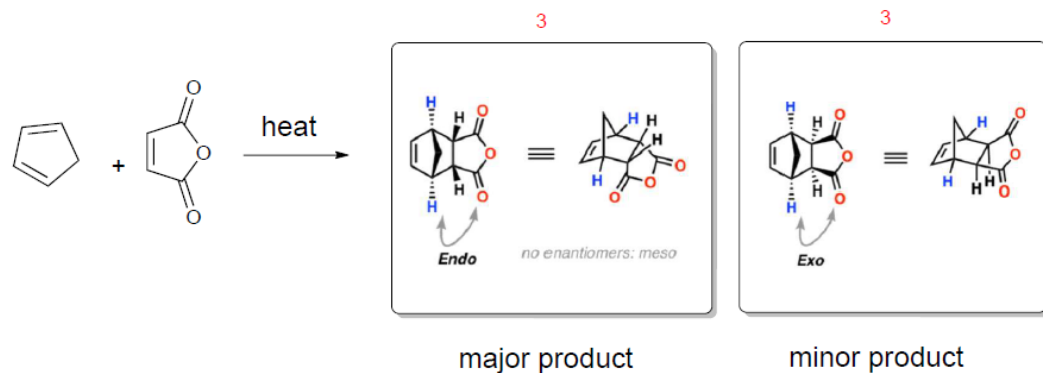
Choice of right chemical reactions enabled synthesis of medicines.



Replacement of imine/enamine and deprotection

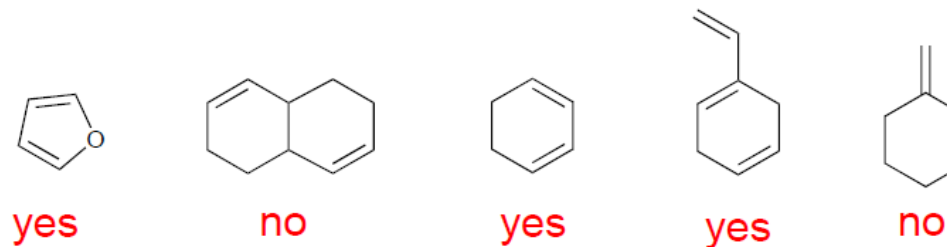
Exam1: Diels-Alder

1. a) predict the products from the following reaction:

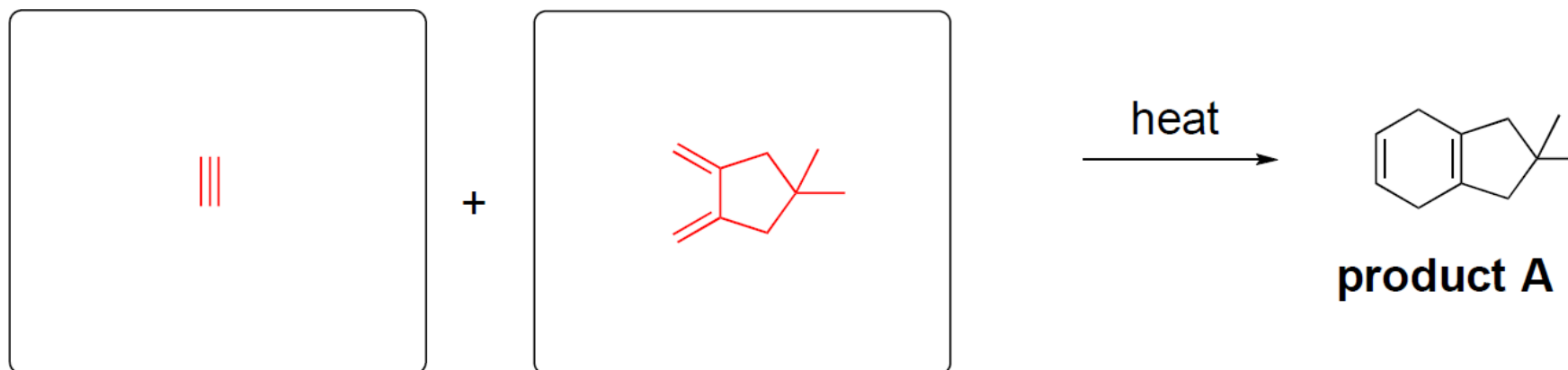


Either depictions are acceptable.

b) Determine if the following molecules are reactive to a dienophile by writing *yes* or *no* under each structure.



10pt (2pt each)

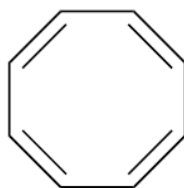


Exam1: Bromination

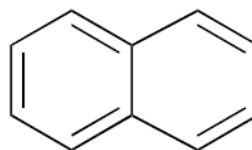
b) Determine if the following molecules are reactive to Br₂ without an activating agent (e.g. Lewis acid) by writing *yes* or *no* under each structure.



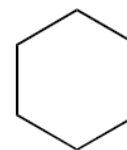
yes



yes



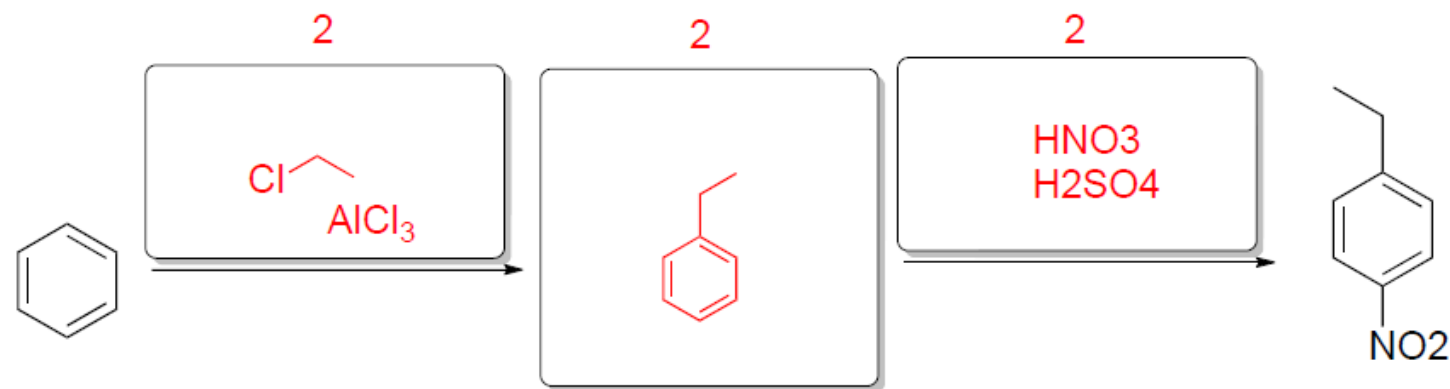
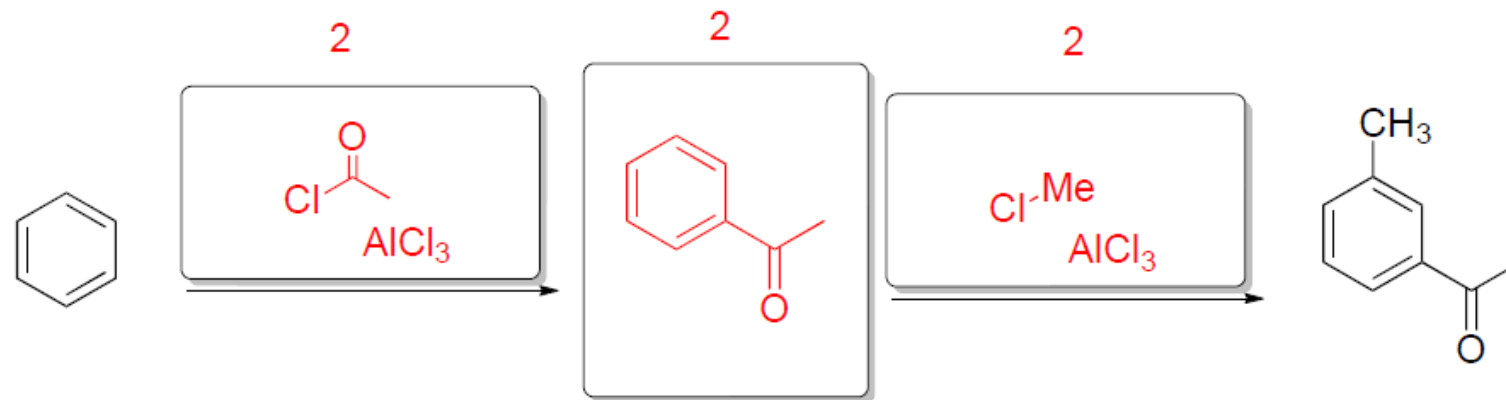
no



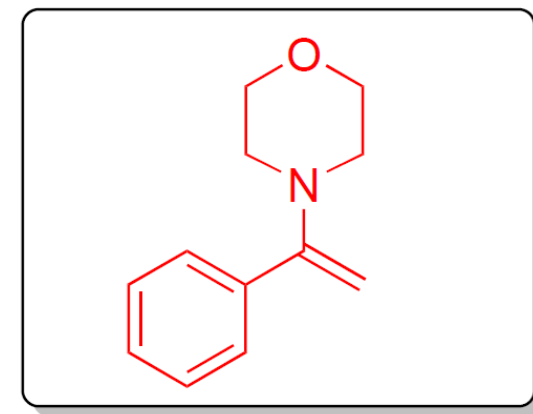
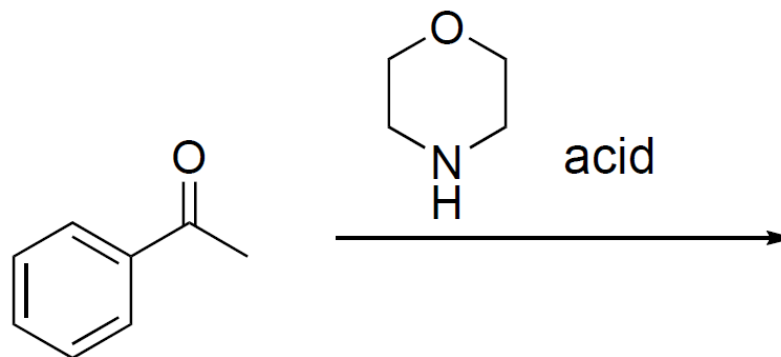
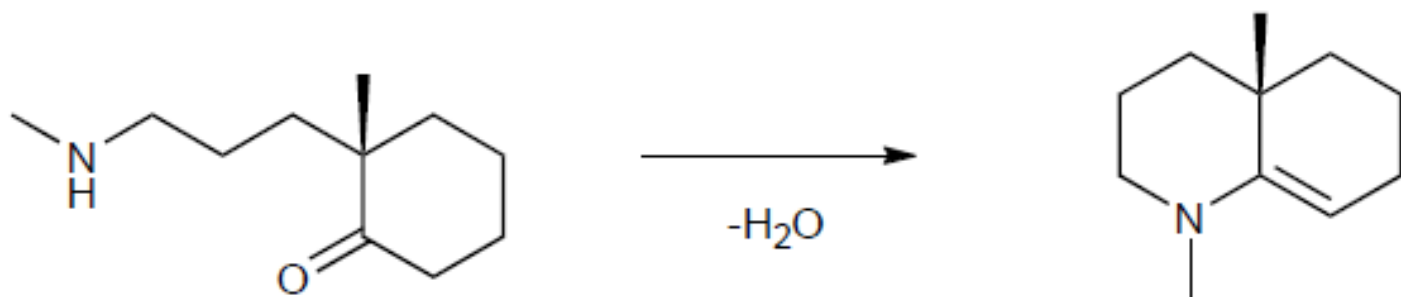
no

4pt (1pt each)

Exam1: EAS order

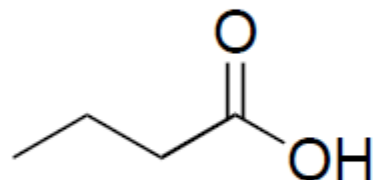


Exam2: Enamine formation

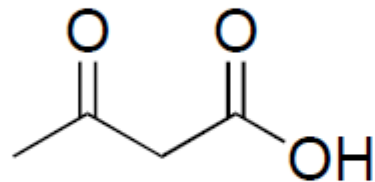


Exam2: Decarboxylation

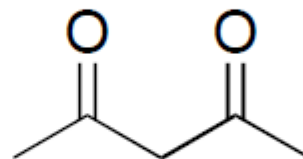
b) Determine if the following molecules can undergo decarboxylation by writing *yes* or *no* under each structure.



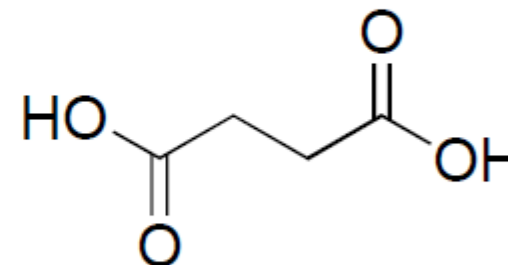
No



Yes



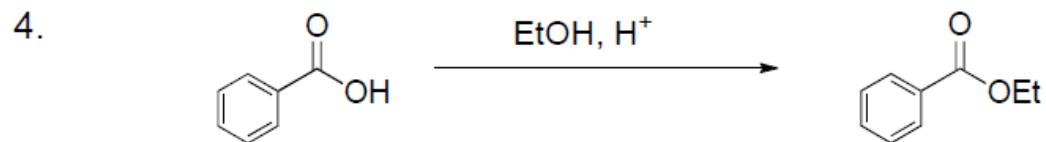
No



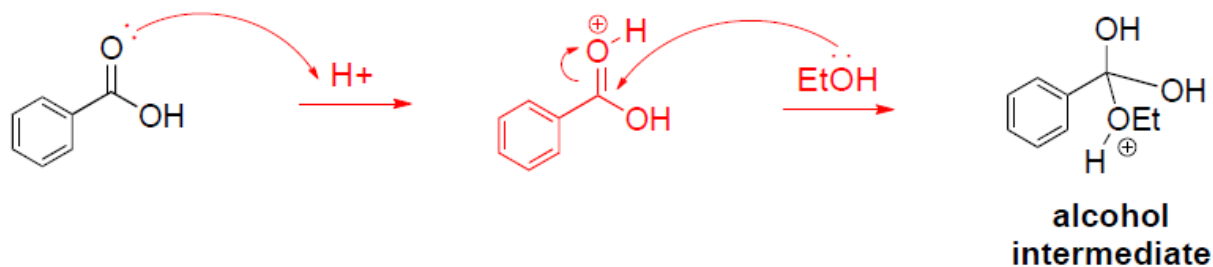
No

2pts each

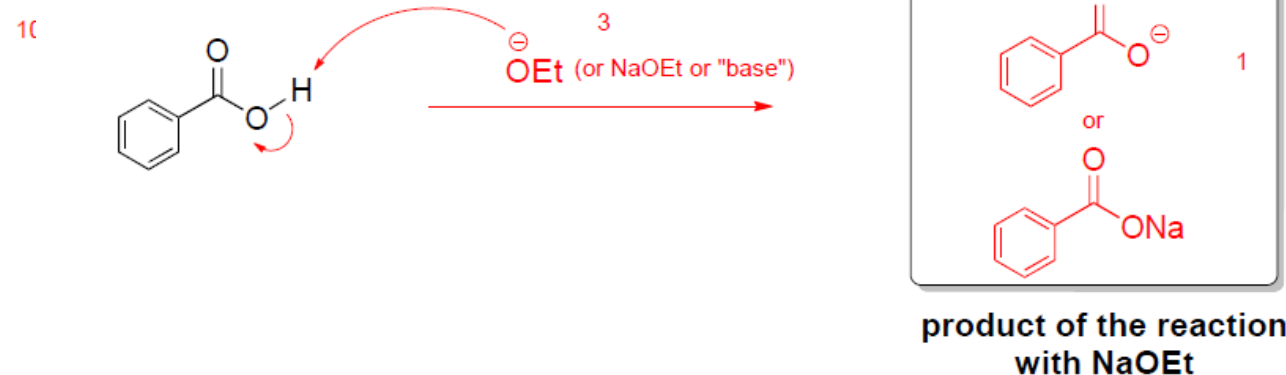
Exam2: Esterification



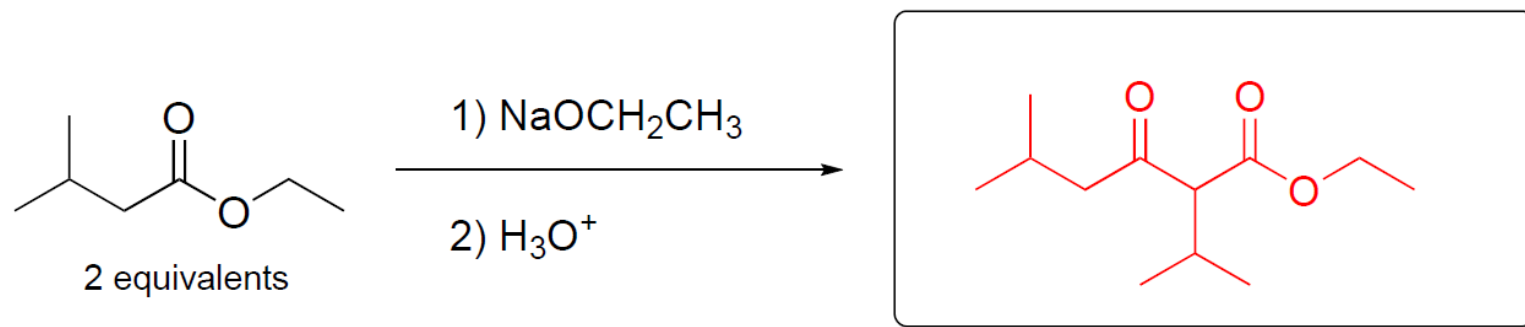
a) Fisher esterification is an acid-catalyzed process. Draw the arrow-pushing mechanism to produce the alcohol intermediate shown below.



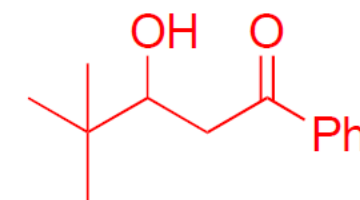
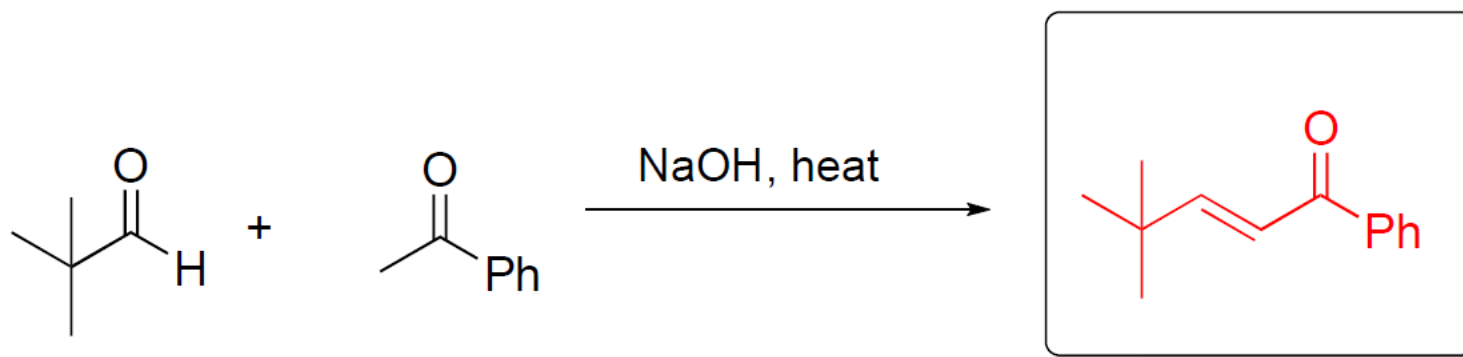
b) On the other hand, a base such as NaOEt (instead of acid) cannot catalyze this reaction. Draw the arrow-pushing mechanism and product structure of a reaction between carboxylic acid and base.



Exam3: Aldol or Claisen condensation



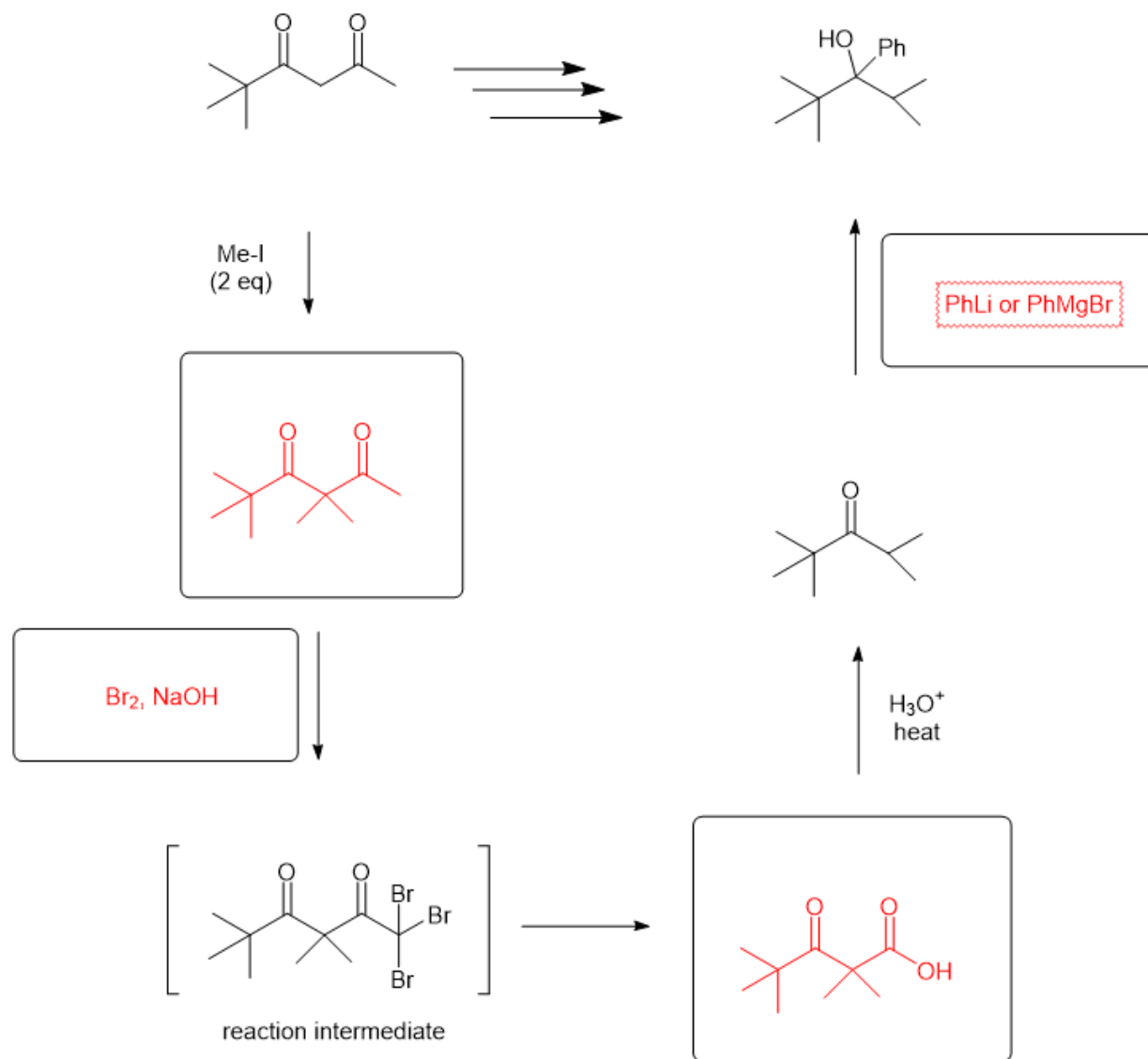
Jun' recommendation: Those are easy to draw wrong product structures. If you've got extra minutes at the end, draw mechanism somewhere else to double check the product structure.



aldol addition
product full points

Chemistry Coach Chapter 8, exercise 1

Exam3: Multistep synthesis



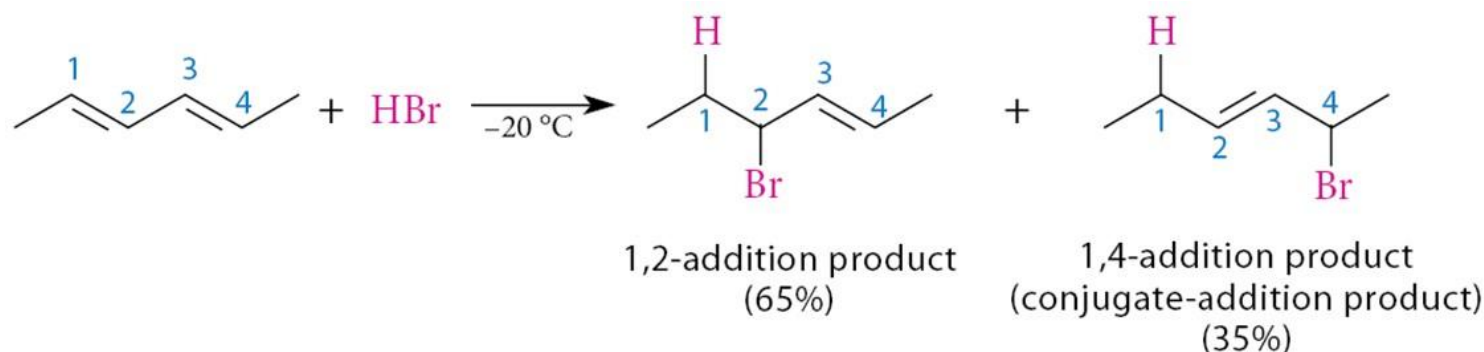
The following slides are highlights from the course.

Chapter 15-17: diene, allylic species, and aromatic comp.

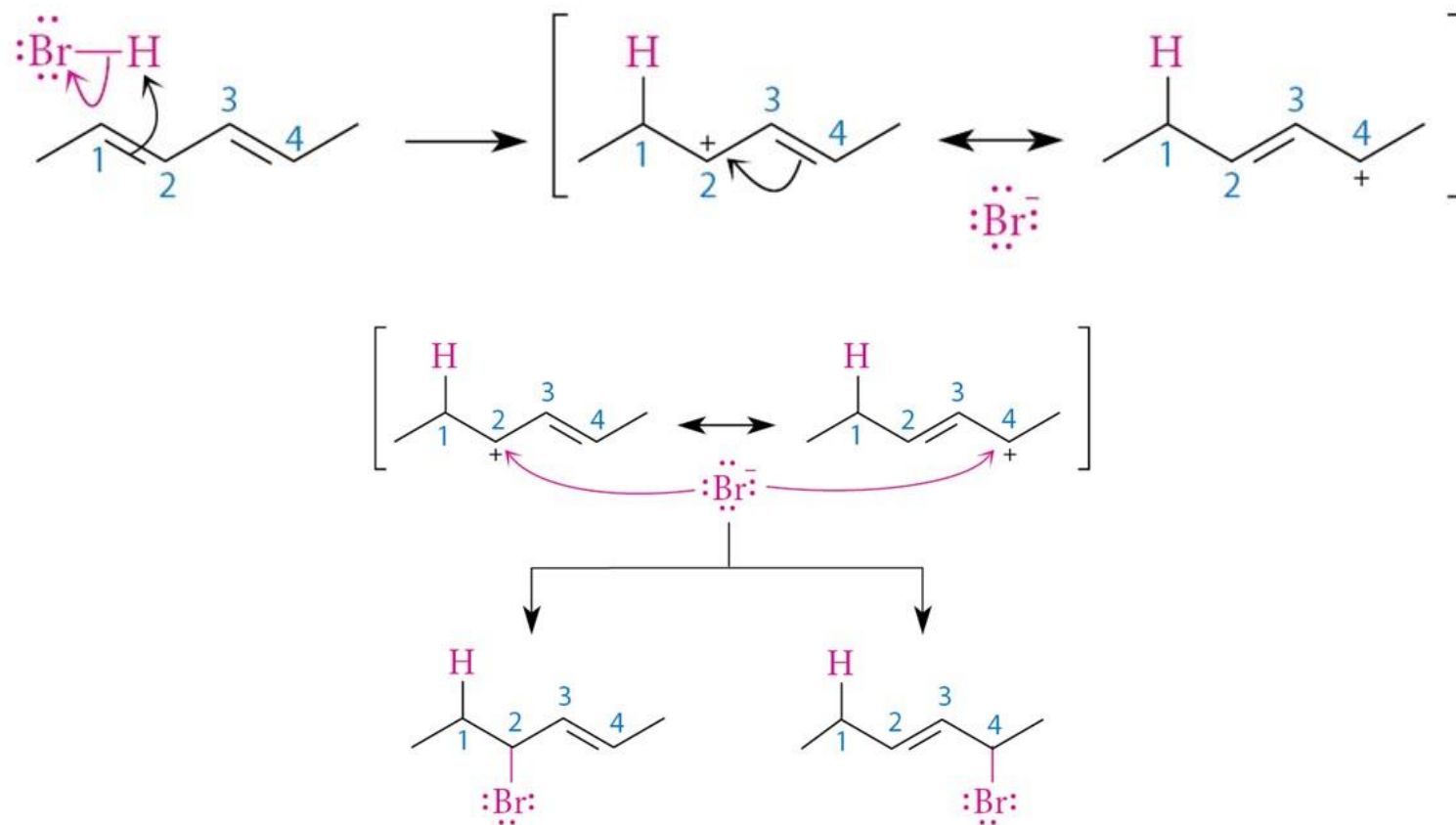
Adding Hydrogen Halides to Dienes: 1,2- and 1,4-Additions

Two types of additions can occur when conjugated dienes react with hydrogen halides:

- **1,2-addition:** Occurs at adjacent carbons.
- **1,4-addition:** Occurs to carbons that have a 1,4-relationship.



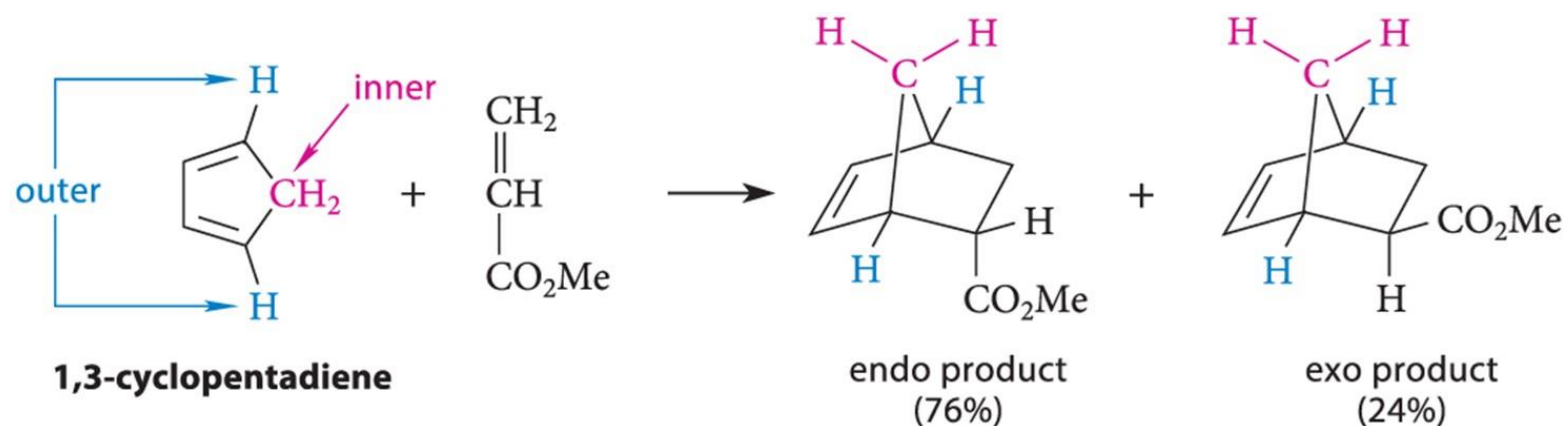
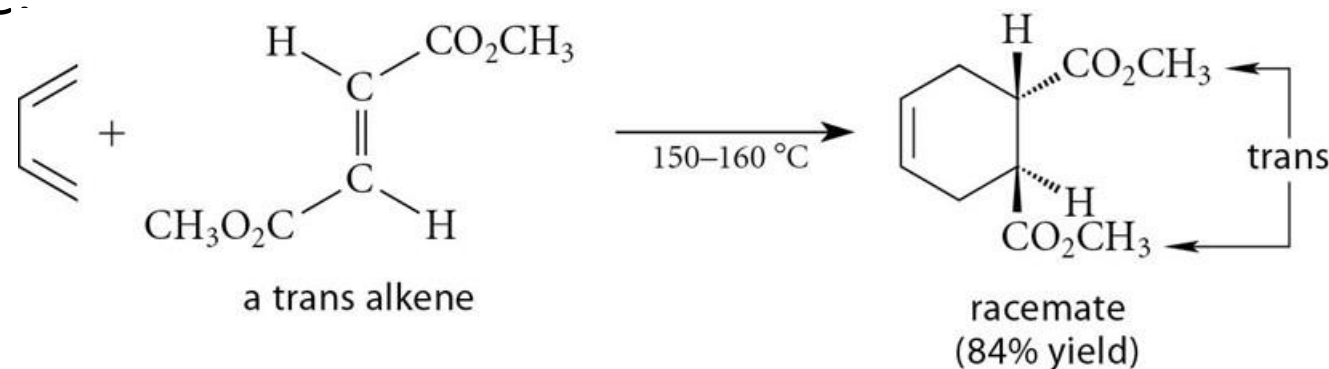
Protonation at C1 of Diene Gives a Resonance-Stabilized Carbocation Intermediate



15.4 Addition of Hydrogen Halides to Conjugated Dienes

Diels–Alder Reaction recap

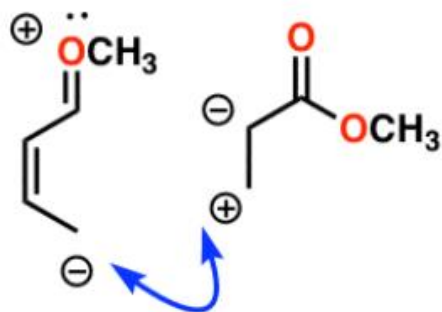
- The concerted transition state results in syn-addition of the diene to the dienophile.



Regioselectivity of Diels-Alder reaction

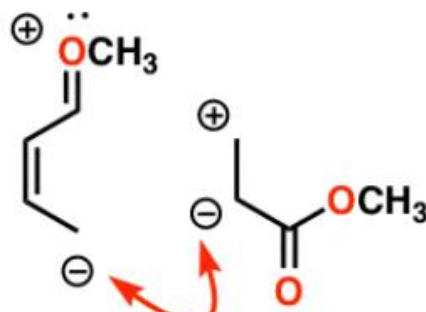
Now, line up the **negative charge from the diene** with the **positive charge on the dienophile**

this way:



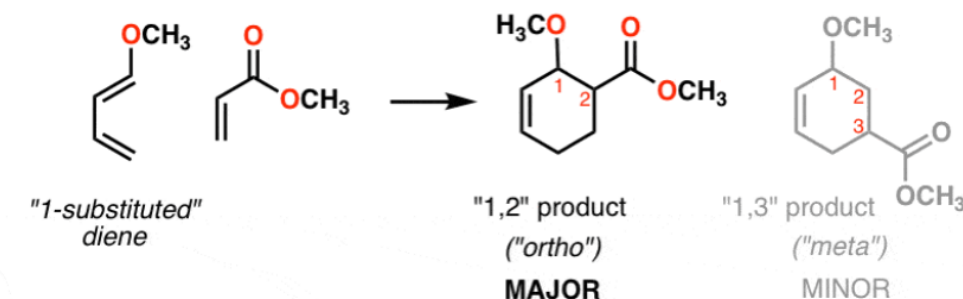
line up opposite charges
results in *ortho*

not this way:



don't line up "like" charges
results in *meta*

Case 1: "1-substituted" dienes: give the "*ortho*" product, not the "*meta*"



<https://www.masterorganicchemistry.com/2018/11/05/regiochemistry-in-the-diels-alder-reaction/>

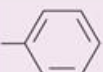
Electronic properties and regioselectivity

- Electron-donating groups (often atom with a lone pair directly attached to benzene): ortho and para directing.
- (exception of halogen)
- Electron-withdrawing groups (acceptor): meta directing.
- Often, atom with multiple bond attached to benzene (no available lone pairs)

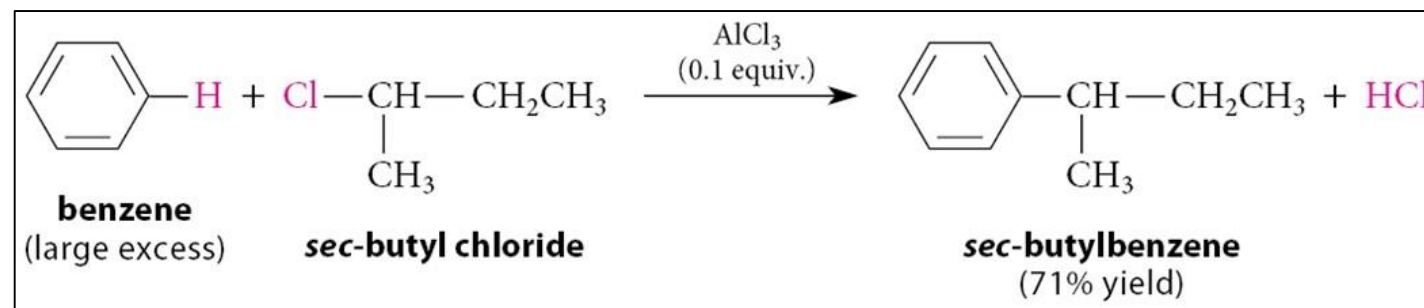
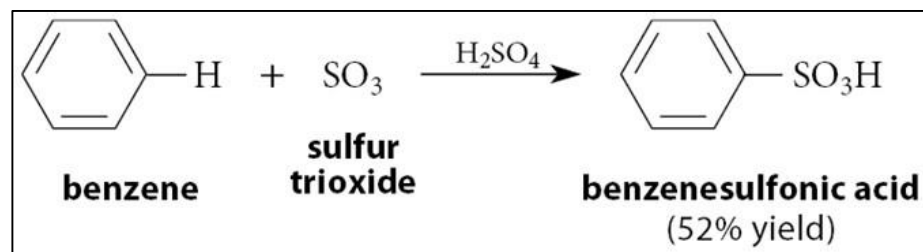
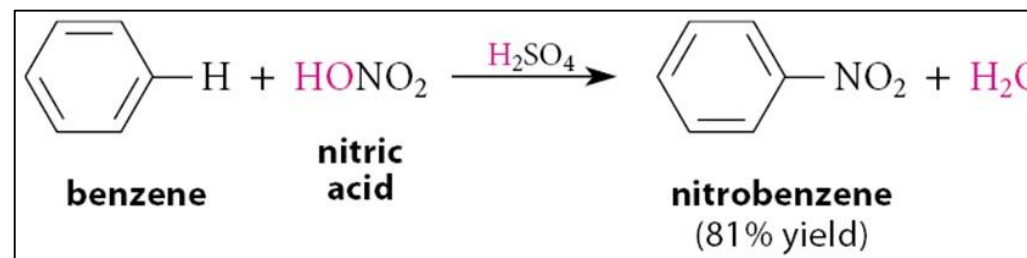
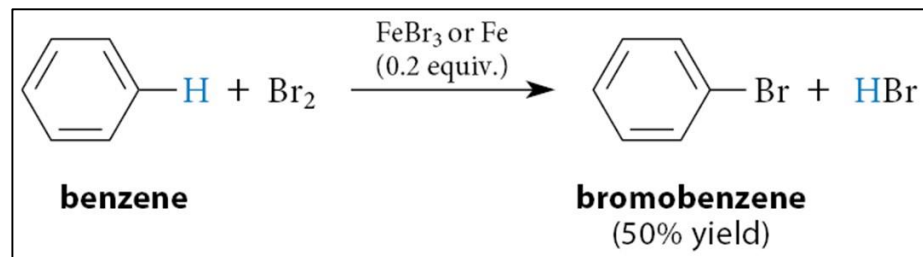
You may look at the periodic table during the final!

Electron-donating groups (EDG)

Electron-withdrawing groups (EWG).

Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$\left\{ \begin{array}{l} -\text{C}(=\text{O})\text{OH} \\ -\text{C}(=\text{O})\text{NH}_2 \\ -\text{C}(=\text{O})\text{OR} \end{array} \right\}$	carboxy, carboxamido, carboalkoxy	meta directors	deactivating substituents
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Introducing Electrophilic Aromatic Substitution

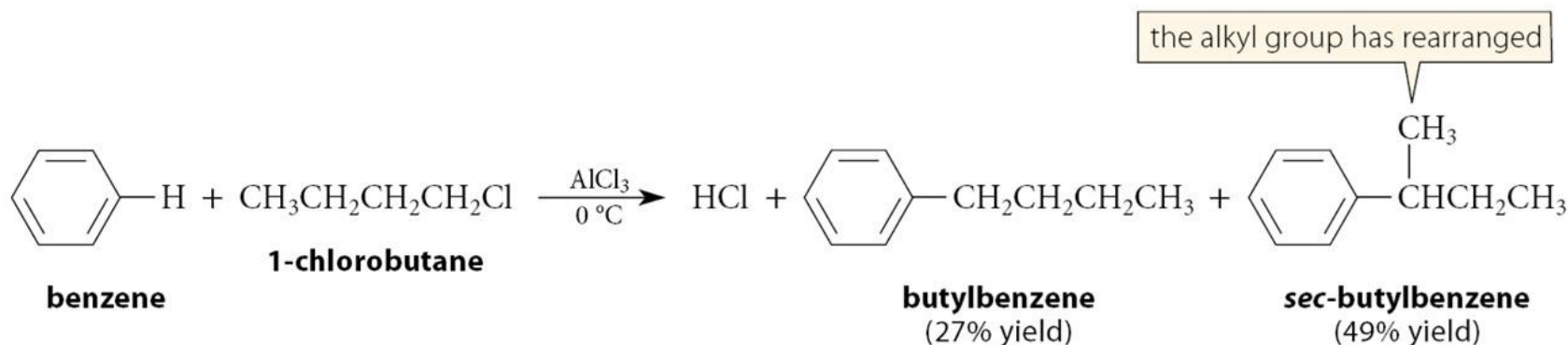


- All electrophilic aromatic substitution reactions occur by similar mechanisms:
Generation of reactive electrophiles attacked by nucleophilic benzene derivatives.

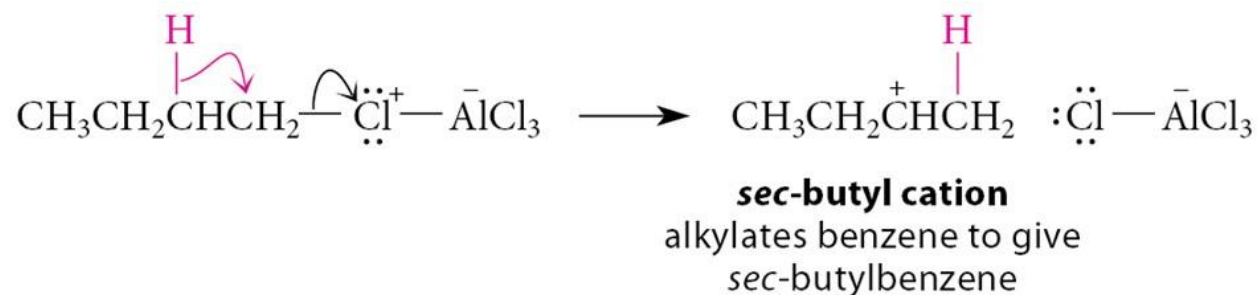
16.4 Electrophilic Aromatic Substitution Reactions of Benzene

Carbocation Rearrangements in Friedel–Crafts Alkylations

- Rearrangements can occur in a Friedel–Crafts alkylation.

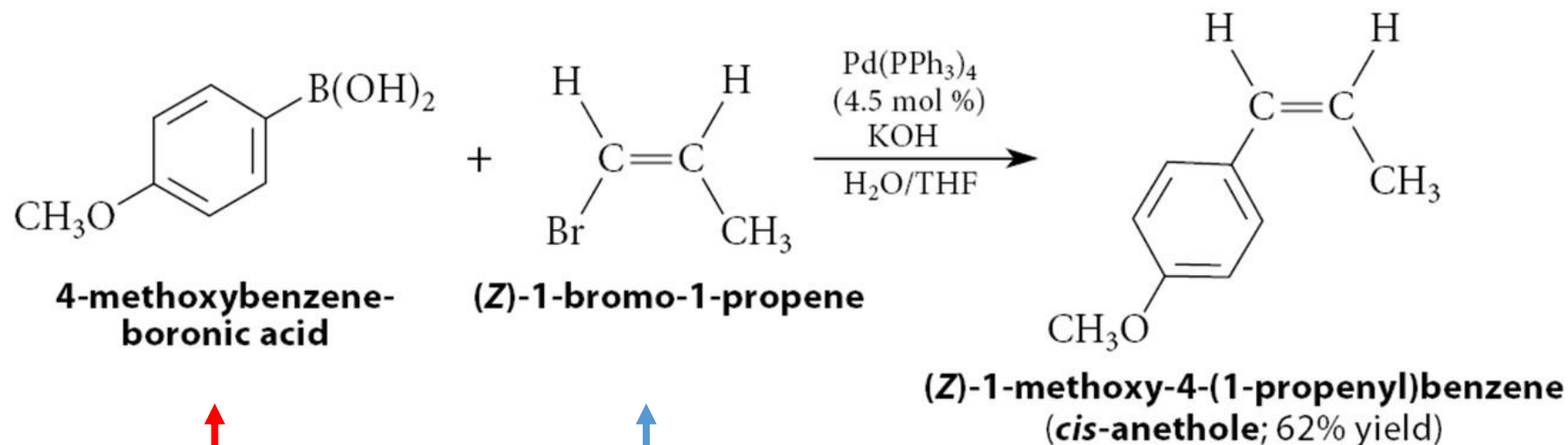


- Primary alkyl halides may rearrange in the AlCl_3 complex stage.



Chapter 18: metal catalysis

Palladium cross coupling looks like SN2 reaction



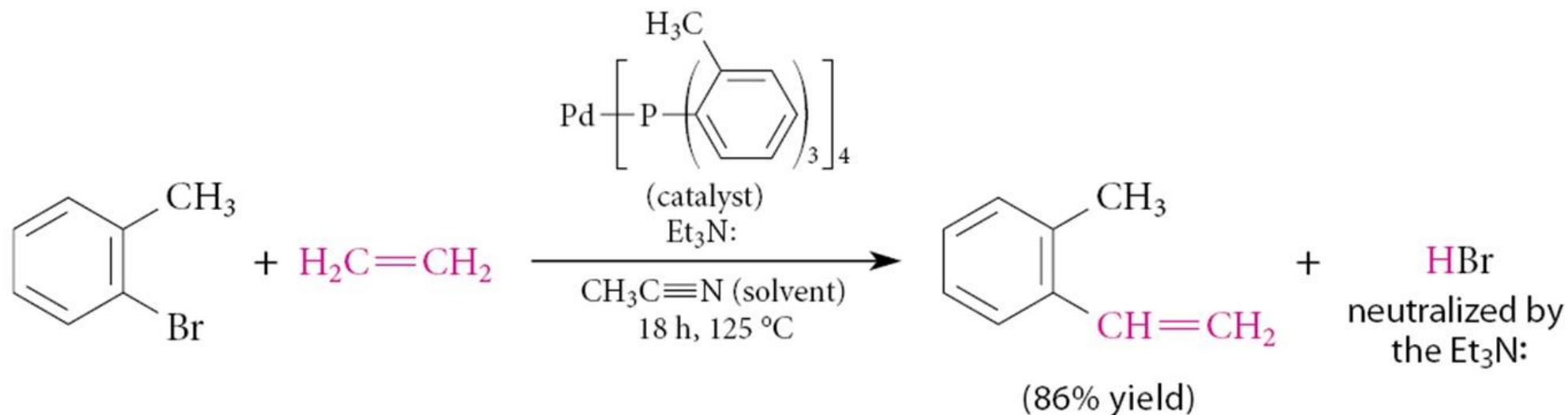
Behave like a nucleophile
(anionic borate species)

Behave like an electrophile

Though SN2 reaction does not happen on vinyl or aryl halides, palladium catalysts virtually makes it possible. (this one is Suzuki coupling)

(Mizoroki-)Heck coupling

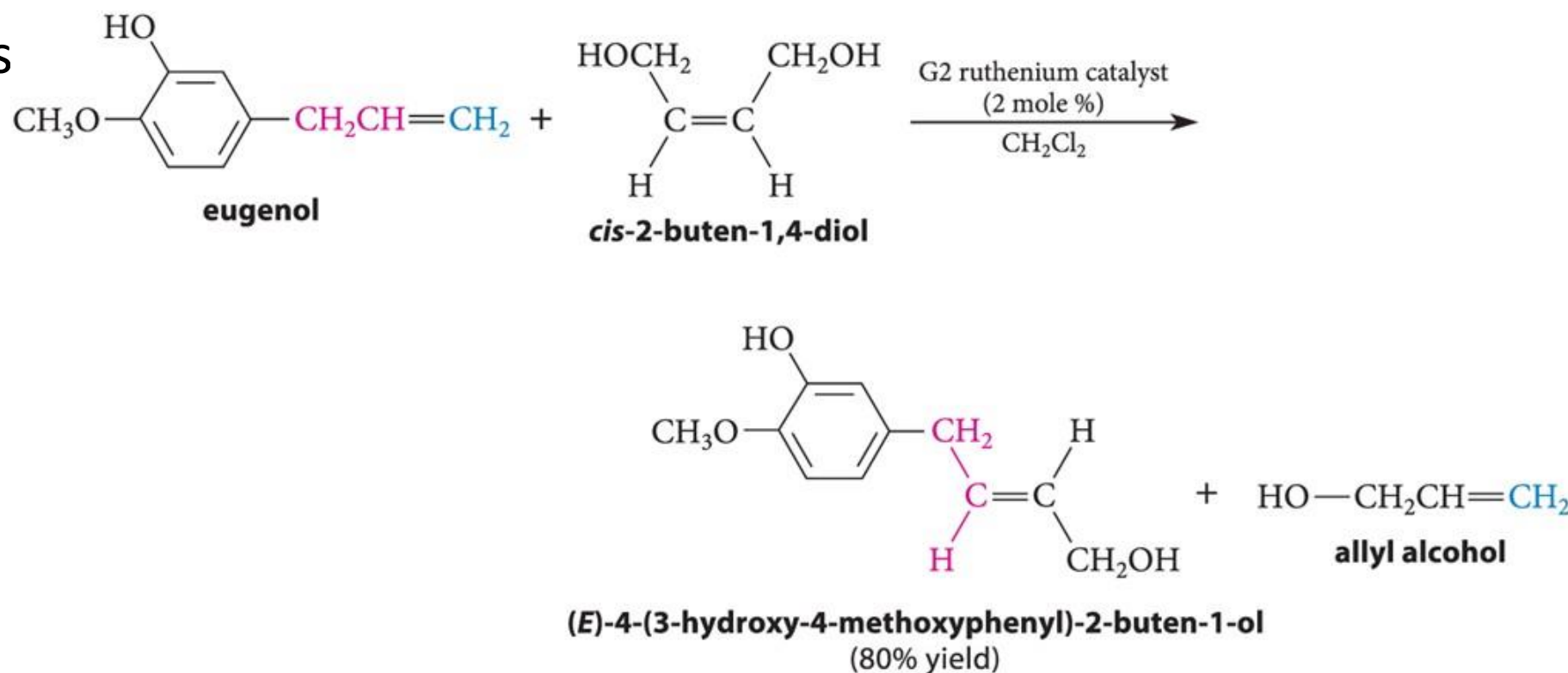
- An alkene is coupled to an aryl bromide or aryl iodide under the influence of a Pd(0) catalyst.



- Slightly different mechanism from Suzuki coupling (i.e. alkene is not behaving like a regular nucleophile. Details on the board)

Alkene Metathesis (Ru catalyst)

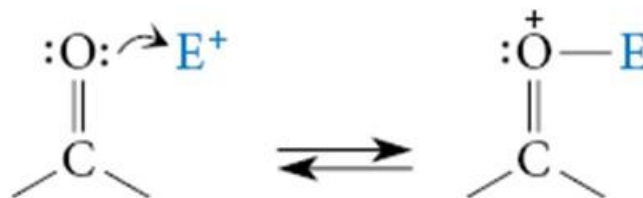
- In an ***alkene metathesis*** the groups at each end of the double bonds are interchanged.
- aka *olefin* metathesis



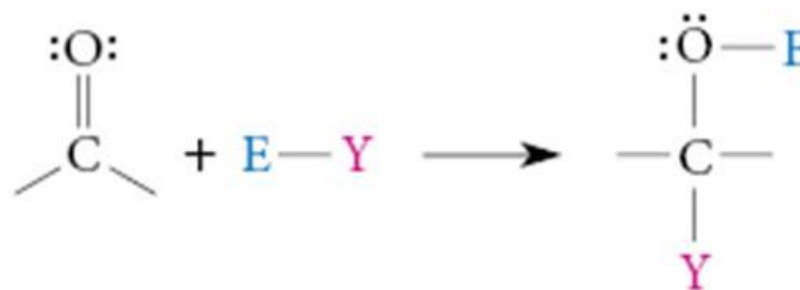
Chapter 19-22: carbonyl chemistry

Chapter 19: Aldehyde and ketone: direct addition

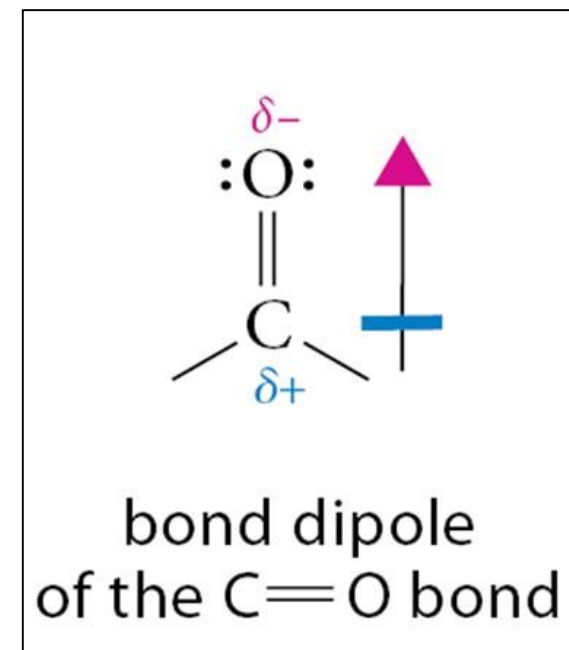
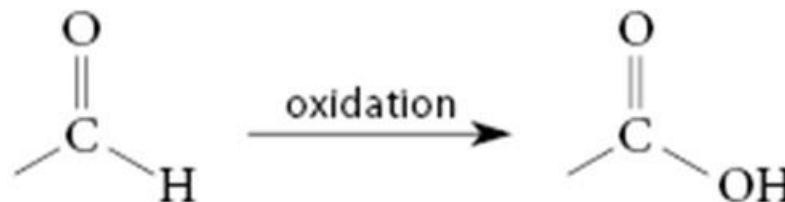
- Reactions with acids



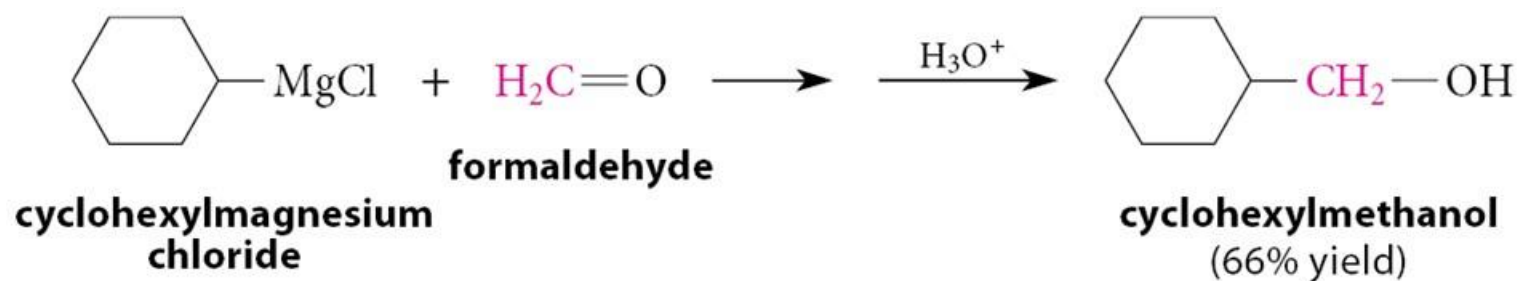
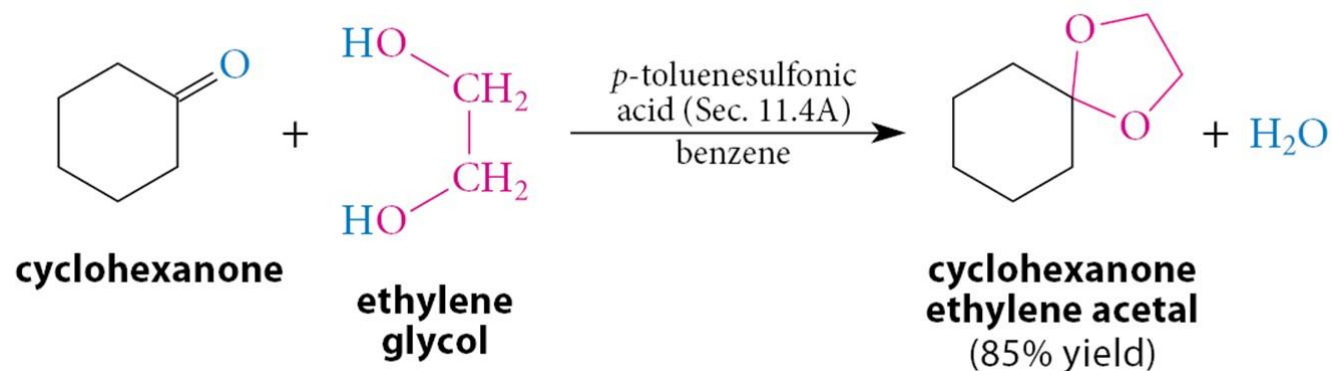
- Addition reactions



- Oxidation of aldehydes

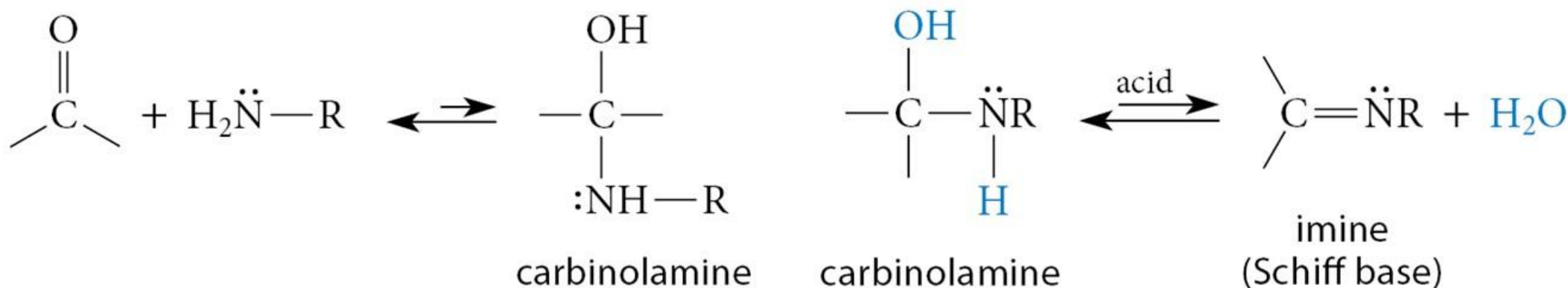


Typical Reactions of Aldehydes and Ketones: Addition Reactions Across C=O

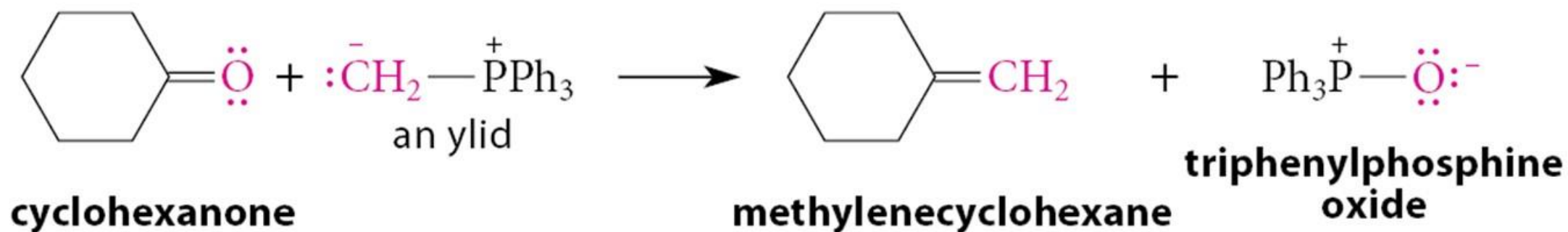


Aldehyde/ketone reactions

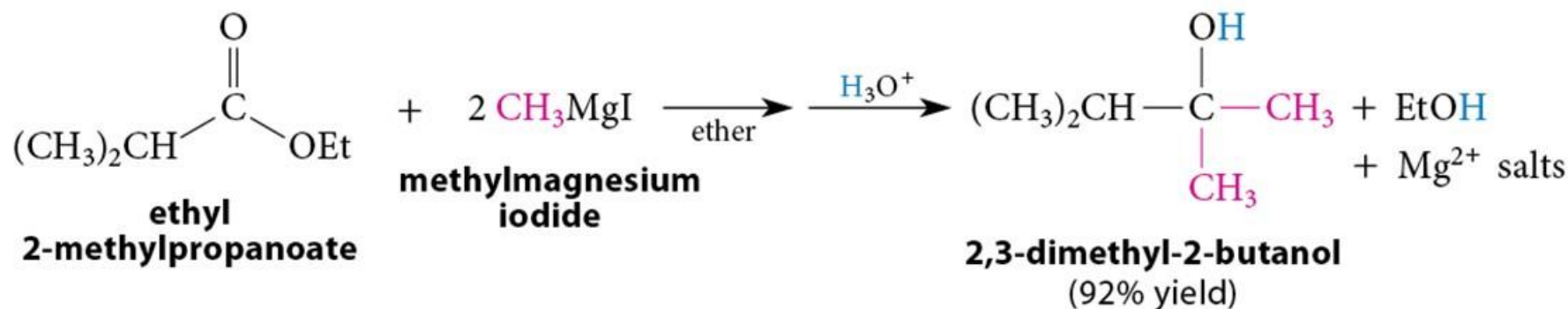
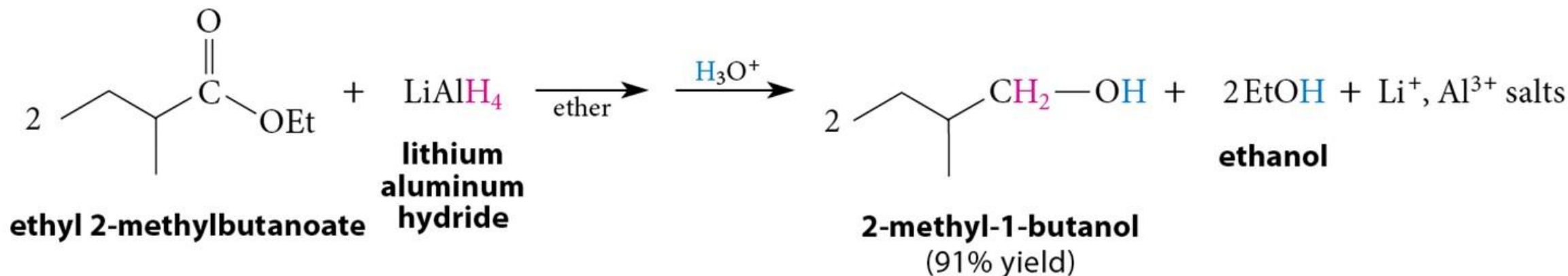
- Imine formation



- Wittig reaction



Chapter 20-21: COOH and its derivatives



- Hydride reduction and hydrolysis of other COOH derivatives should be learned too (they are quite similar though)

Hydrolysis of Esters

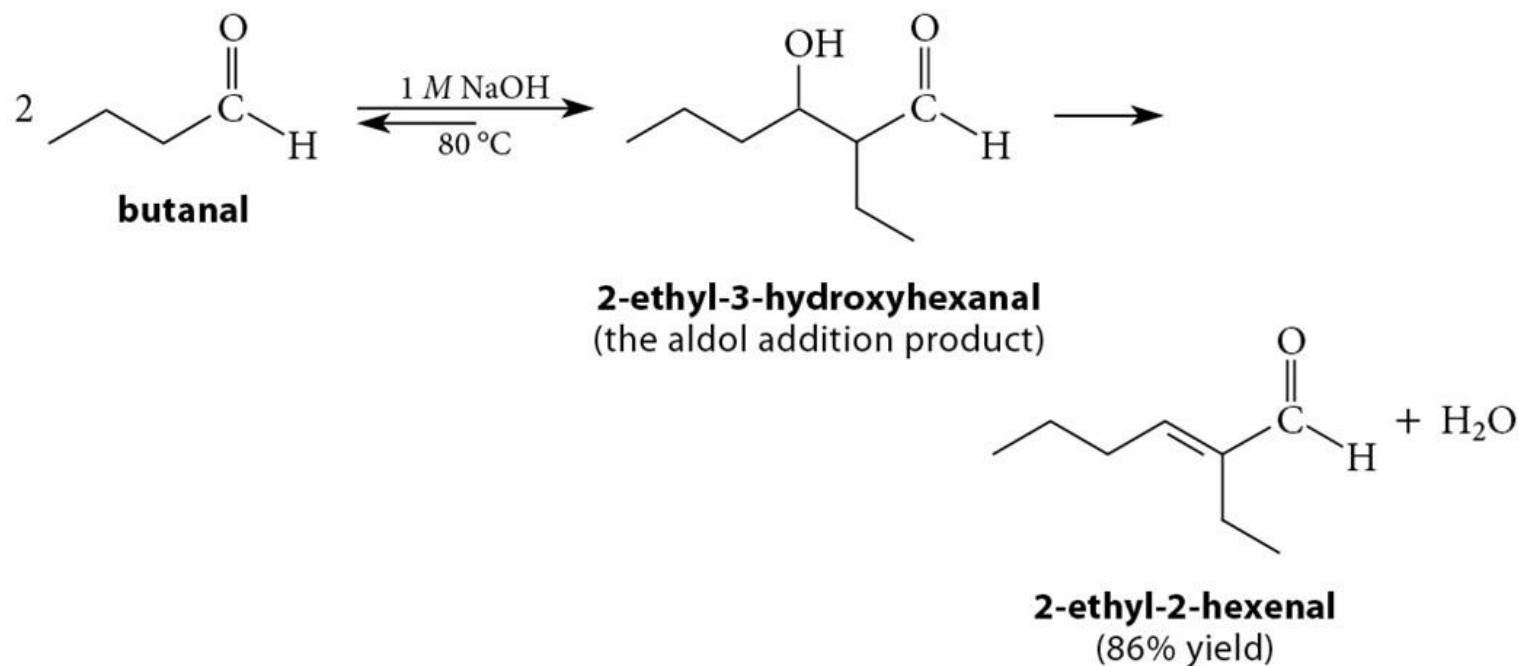
- Saponification* – ester hydrolysis in aqueous hydroxide.



- Make sure to differentiate the acid- vs base-catalyzed mechanisms!**

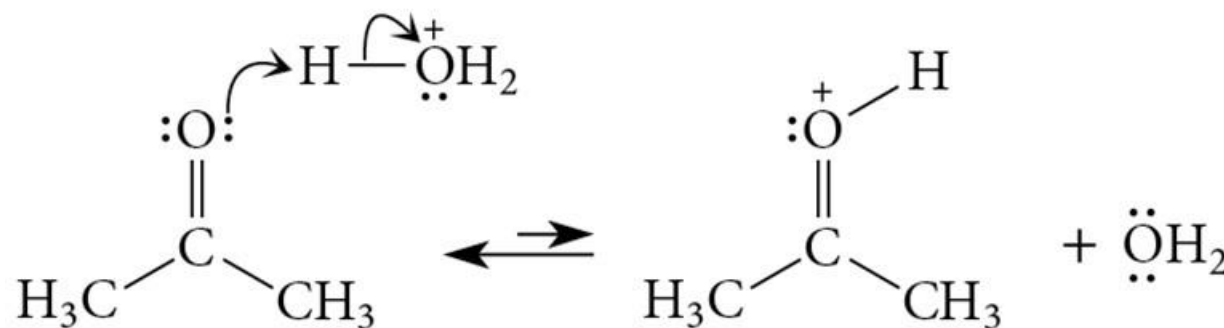
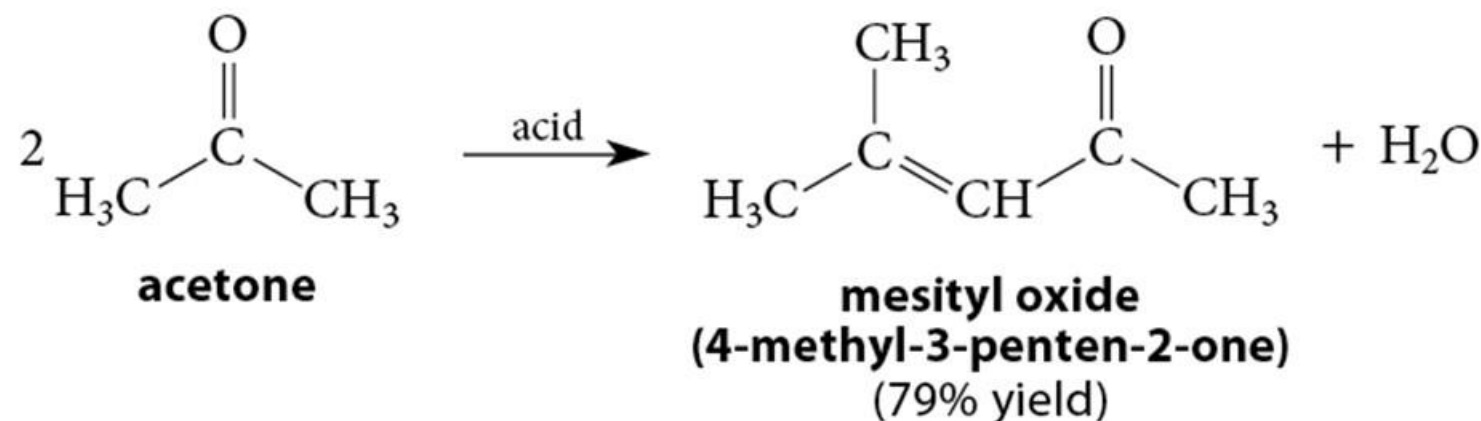
Chapter 22: Enolate. Aldol Addition + Dehydration = Aldol Condensation

- Under more severe conditions (higher base concentration or higher temperature), the aldol product can undergo a *dehydration reaction*.



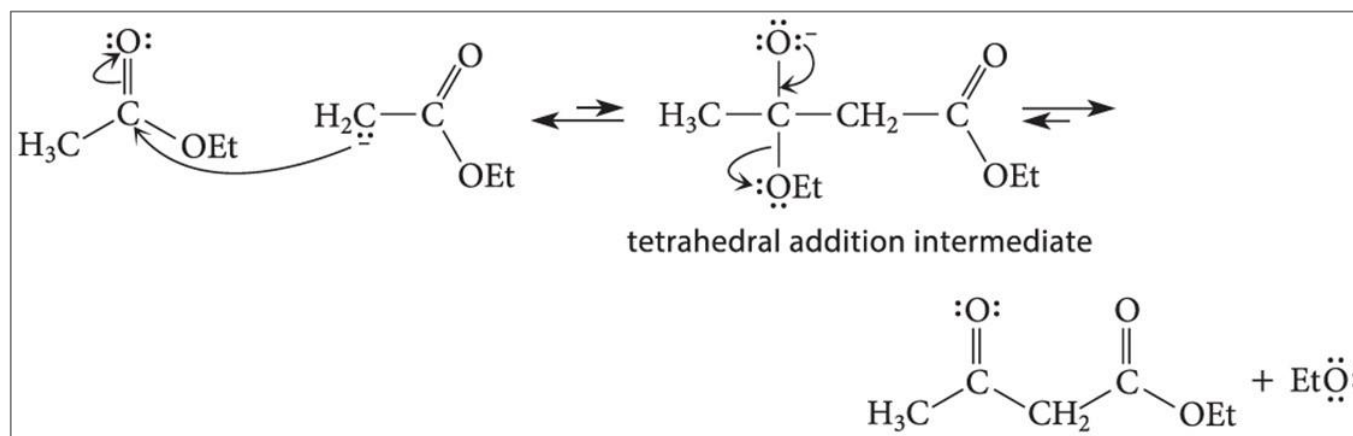
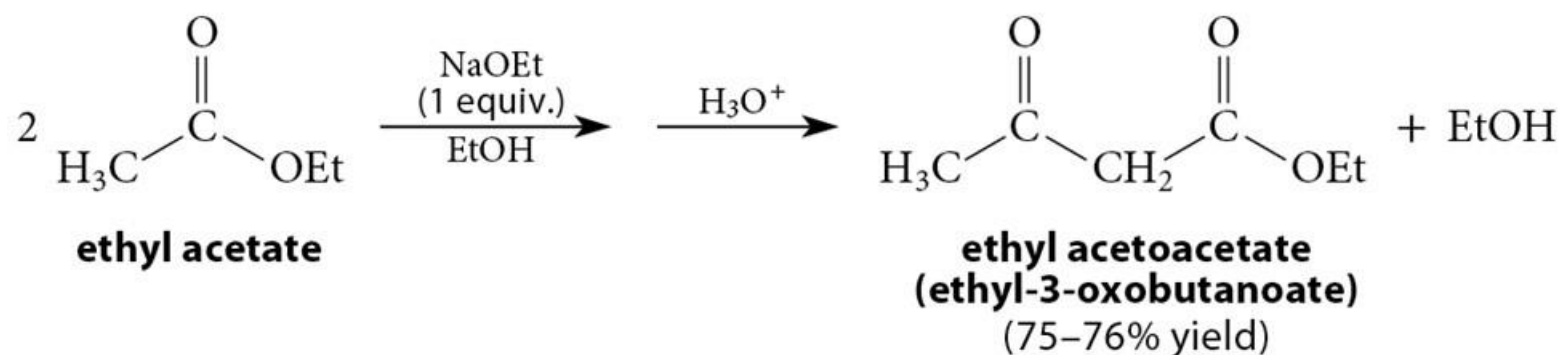
Acid-Catalyzed Aldol Condensation

- With acid, aldol *condensation products* are formed; addition products cannot be isolated.
- Protonated aldehyde/ketone is key intermediate.



Claisen Condensation (like aldol but with esters)

- A condensation reaction involving *esters*.

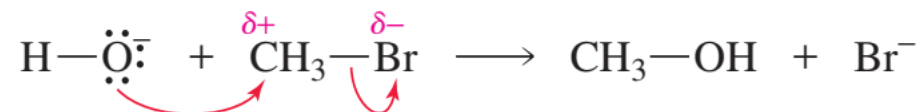


22.6 Condensation Reactions Involving Ester Enolate Ions

Three typical reaction types in organic reactions

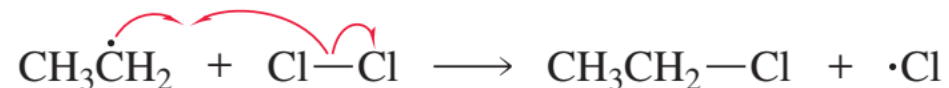
- A **polar reaction** is one in which a nucleophile reacts with an electrophile. Both electrons in the new bond come from the nucleophile. Most of the reactions we saw in previous chapters are polar reactions.

a polar reaction



- A **radical reaction** is one in which a new bond is formed using one electron from each of the reactants.

a radical reaction



- A **pericyclic reaction** is one in which the electrons in one or more reactants are reorganized in a cyclic manner. The only pericyclic reaction we have seen so far is the Diels–Alder reaction.

- Note that metal-related chemistry (inorganic/organometallic) could be different from those.